

Incident Assessment & Financial Exposure Report

Generated by 5-Agent LangGraph Orchestrator & Risk Engine

Incident ID:	INC-E3A9C8D4	Assessment Date:	2026-07-11 00:30:15 UTC
Target Facility:	Power Plant (device-02)	Location:	Zone-B / Grid East
Attack Signature:	Denial of Service (DoS)	Incident Severity:	CRITICAL
Containment State:	ISOLATED (Network Quarantine)	Isolation Port:	MQTT Channel 1883

1. Technical Incident Narrative

At approximately 00:30:15 UTC, the telemetry gateway flagged critical sensor anomalies from Device 02 (Power Plant). The **Detector Agent** running local Random Forest Classifier computed an anomaly score of **0.941**, indicating a high-confidence attack vector. The state was dispatched to the **Classifier Agent** which analyzed telemetry signature characteristics: temperature spiked to 120°C and pressure breached normal thresholds at 380bar, which is a classic signature for a Denial of Service (DoS) load attack on turbine control units.

The **Isolator Agent** triggered network micro-segmentation commands, issuing a quarantine payload to the MQTT broker to isolate Device 02 from the main industrial network. This containment action successfully prevented lateral movement to nearby SCADA equipment.

2. Financial Blast-Radius Quantification

The **Risk Quantifier Agent** mapped the technical parameters (downtime duration and device type) into a structured economic loss model. Because the power plant has a critical grid-stabilization SLA, downstream penalties accumulate rapidly. Fines were calculated according to NERC CIP regulatory enforcement schedules.

Cost Component	Calculation Method / Rate	Estimated Loss (USD)
Downtime Operating Losses	8.0 Hours @ \$500,000.00 / Hour	\$4,000,000.00
SLA Grid Penalty	15% of Downtime Cost	\$600,000.00
NERC CIP Regulatory Fine	Severity Level 2 Schedule Lookup	\$2,000,000.00
Total Exposure	Sum of Losses	\$6,600,000.00

3. Bank Credit Risk Assessment

CREDIT RISK ALERT: CRITICAL RISK FLAG

The estimated total exposure of \$6,600,000 breaches the lender's \$2,000,000 warning threshold.

Recommendation: Initiate immediate covenant review. The operator of the facility exhibits elevated debt service risk due to expected working capital depletion. Verify operator's cybersecurity liability insurance coverage and check if the isolated device has been restored via secondary secure bootstrap.

Report verified by Aegis Reporter Agent. Compliance standard: NIST SP 800-61.